Correlated translated probes and the bound $\Lambda < 0.149396$ for the de Bruijn–Newman constant

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Abstract

A computer-assisted argument gives the upper bound $\Lambda \leq 0.149395073827688 < 0.149396$ for the de Bruijn–Newman constant in the Polymath normalisation. The construction starts from the classical width-one zero strip at time zero. A finite-head localisation wall and uniform logarithmic-derivative estimates are coupled to a decreasing zero-height barrier. The main refinement retains nonnegative surplus terms in a one-point value–derivative inequality: their phases at two prescribed translations cannot simultaneously realise the separate worst cases. A second gain comes from the finite Euler logarithmic derivative. The proof accommodates unequal coefficients and denominators, source-frequency seams, physical cutoff changes and the complete approximation remainder. The source certificates cover 608 time cells, with six frequency ranges per cell including an infinite final range. The terminal comparison is enclosed by exact rational arithmetic. The deposited source and evidence allow the numerical inequalities to be regenerated. The analytic arguments, including the larger-height approximation, are provided in a companion text; they are not formalised by the numerical replay.

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1 Introduction

The de Bruijn–Newman constant Λ is the threshold at which every zero of the heat-deformed Riemann xi function is real. In the normalisation used here, the Riemann hypothesis is equivalent to $\Lambda \leq 0$; the theorem of Rodgers and Tao gives $\Lambda \geq 0$ [1, 3]. The present paper concerns an explicit positive upper bound, not the zero endpoint.

Previous and concurrent bounds. Polymath proved $\Lambda \leq 0.22$ [1]. Platt and Trudgian's finite verification of RH subsequently gave $\Lambda \leq 0.2$ [2, Corollary 2]. Among the subsequent public preprints, Gomila announced $\Lambda \leq 893927/5000000 = 0.1787854$ in August 2026 [4], and the author's earlier Zenodo paper reported $\Lambda < 0.163297$ in September 2026 [5]. Gordon's concurrent September 2026 manuscript reports $\Lambda \leq B_G = 0.157656619095490606768\ldots$ [6], where

$$B_G = \frac{3885632262767861213460393068710302759}{24646172707879668706230182733520000000}.$$

Gordon also starts from the classical strip at time zero, but uses different density and jet estimates and a partial Lean formalisation whose deduction retains explicit analytic hypotheses. The recent

preprints are cited for context, not as proof dependencies. No theorem, estimate or numerical certificate from Gomila's or Gordon's work is used to obtain the bound below; the required new estimates and computations are supplied here. This statement concerns the mathematical dependency of the proof, not a historical priority claim. The comparison sources and their status were checked on 21 September 2026; no exhaustive survey of priority is asserted.

Theorem 1 (Main result)**.** *For the de Bruijn–Newman constant in the normalisation of Section 2,*

$$\Lambda \leq \frac{18674384228461}{125000000000000} = 0.149395073827688 < 0.149396.$$

The earlier paper [5] uses a fixed positive-time row and a final-time covering. Neither its narrow-strip theorem nor its atlas is assumed here. Instead, the argument combines a time-zero localisation wall, all-frequency one-point bounds, a translated-phase surplus, and a continuous zero-height comparison. The contribution is the quantitative two-probe improvement with its complete physical and arithmetic accounting, rather than a new proof of the classical heat-flow framework.

1.1 Dependencies and verification scope

The companion *Analytic Foundations* is part of the mathematical submission. It reproduces the supporting source derivations at their exact scope. The following map distinguishes published inputs, new analytic arguments and finite numerical certificates.

Ingredient	Precise source	Role
Initial strip; dynamics; collisions; confinement	[1, Proposition 3.1; Theorems 3.2 and 1.5(i)]	Published analytic inputs
Finite RH through 3000175332800	[2, Theorem 1]; Foundations A, Section 2	Published computational theorem
Localisation wall	Foundations A, Section 3; regenerated prefix inequalities	New finite certificate
Larger-height approximation and derivative errors	Foundations A, Sections 4–5, derived from [1, Section 6]	Written extension and interval bounds
One-point bounds and sharp geometry	Foundations A, Sections 6–12; Foundations B, Sections 1–5	Supporting analytic and numerical results
Correlated gain and final endpoint	Sections 3–8 and 11; deposited certificates	Present refinement

The larger-height approximation is not an unqualified invocation of the height-restricted numerical statement of [1, Theorem 1.3]. Its derivation and complete remainders are in Foundations A, Sections 4–5. Both transcendental calculations use Arb/FLINT [7]; their different precision and construction choices do not make them independent libraries. An additional standard-library checker validates the rational comparison histories. Numerical regeneration checks the finite inequalities, whereas the analytic reductions require mathematical review. This preprint has not undergone external peer review or complete proof-assistant formalisation.

1.2 Organisation and reproducibility

Sections 2–6 construct the joint source estimate. Sections 7–8 apply it to zero motion. Sections 9–11 state the implementation, limitations and exact terminal comparison. `Proof_and_Replay.zip` contains the tested sources, numerical runtime, certificates and default regeneration command. The separate Verification Guide distinguishes file-integrity checking from numerical regeneration and analytic review. The old atlas volumes are not required.

2 The source model and the one-point estimate

Use the same normalisation

$$H_t(z) = \int_0^\infty e^{tu^2}\Phi(u)\cos(zu)\,du, \qquad H_0(z) = \xi((1+iz)/2)/8.$$

Here

$$\Phi(u) = \sum_{n=1}^{\infty}(2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u})e^{-\pi n^2 e^{4u}}, \qquad \xi(s) = \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s).$$

The normalisation, the identity $\partial_t H_t = -\partial_z^2 H_t$, and the classical zero dynamics follow the conventions of [1]. All zero counts include multiplicity. The detailed source maps, normaliser B_t, and branches of the logarithm are given in Analytic Foundations, Part A, Section 4.

Write $S(x,h,t) = -\Im(H_t'/H_t)(x+ih)$. The reference source has $X = 6000000185827$, $N_0 = 690988$, $L = \log(x/(4\pi))$, and

$$s_U^0 = (1+h)/2 + tL/4 - i(x/2+\pi t/8), \qquad s_V^0 = (1-h)/2 + tL/4 + i(x/2+\pi t/8).$$

Its leading gamma is $\gamma_0 = \gamma(x,t,0)e^{-hL/2}$. The finite Euler multiplier is

$$E_t(s) = \prod_{p\in\{2,3,5\}}(1-b_t(p)p^{-s}), \qquad b_t(n) = e^{t\log^2 n/4}.$$

The finite Euler multiplier in this paper is distinct from the connected exponential normalisers used in the earlier fixed-time programme.

Write

$$P_N(s) = \sum_{n\le N} b_t(n)n^{-s}, \qquad N = N(x,t) = \left\lfloor \sqrt{\frac{x}{4\pi} + \frac{t}{16}} \right\rfloor.$$

All logarithms of positive real numbers are natural logarithms. At each actual source, the complete leading core uses this actual cutoff:

$$Q = E_t(s_U^0)(P_N(s_U^0) + \gamma_0 P_N(s_V^0)) = \sum_{j\ge 0} a_j z_j,$$

with $a_0 = 1, z_0 = 1$, all $a_j \ge 0$, and $|z_j| = 1$. The product head through N_0 is combined first; the full product tail, reflected branch and every later activated term remain explicit signed labels (their signs are absorbed in their phases). At fixed x, t, N,

$$Q_h = -V/2, \qquad V = \sum_j \lambda_j a_j z_j, \quad \lambda_0 = 0, \quad \lambda_j \ge 0.$$

For first-head index n, $\lambda_n = \log n$ and

$$z_n = \exp\big(i(x/2+\pi t/8)\log n\big).$$

For reflected label (d,n), $\lambda = L - \log n + \log d$.

The one-point calculation in Analytic Foundations, Part A, certifies, on each complete time–frequency rectangle,

$$0 \le A_- \le A := \sum_{j>0} a_j \le A_+ < 4/5, \qquad B := \sum_j \lambda_j a_j \le B_+,$$

with a rational $C \in (0, 3.4)$ for which

$$(1+A_-)[C(1+A_-) - B_+] - D_+ > 0,$$

where D_+ bounds the complete negative-pair budget defined below. It also provides the Euler logarithmic-derivative cap E_ℓ, the complete value and height-derivative remainder caps R_0, R_1, and a strict physical floor ℓ. These are reconstructed, not read solely from a PASS marker.

The physical transfer in Analytic Foundations, Part A, Sections 5 and 9, yields

$$S > \ell + \frac{C|Q|^2 - \Re(V\overline{Q})}{2|Q|^2} + \left(E_\ell - \Re \frac{(E_t(s_U^0))_h}{E_t(s_U^0)} \right). \tag{1}$$

Here ℓ is rounded down from

$$L_{\text{lower}}/4 - 10^{-6} - E_\ell - C/2 - \frac{R_1 + B_+ R_0/[2(1-A_+)]}{1 - A_+ - R_0}.$$

Equation (1) is the one-point argument with the discarded nonnegative terms retained. Both complete physical remainders are still charged. Their caps do not depend on the discarded phases. The leading replacement and its height derivative are justified before using (1); no unknown error is differentiated.

3 The nonnegative surplus in the one-point estimate

Set

$$w_{ij} = a_i a_j (\lambda_i + \lambda_j - 2C), \qquad D_C = 2 \sum_{i<j,\ w_{ij}<0} |w_{ij}|.$$

Lemma 2 (Surplus decomposition)**.** *With the notation above, the following identity holds. Each term in its last two sums is nonnegative.*

$$\begin{aligned} H_C &:= C|Q|^2 - \Re(V\overline{Q}) \\ &= (1+A)[C(1+A) - B] - D_C \\ &\quad + \sum_{w_{ij} \geq 0} w_{ij}[1 - \Re(z_i \overline{z}_j)] \\ &\quad + \sum_{w_{ij}<0} |w_{ij}|[1 + \Re(z_i \overline{z}_j)]. \end{aligned} \tag{2}$$

Proof. Expand $|Q|^2$ and $\Re(V\overline{Q})$, group the terms with indices $i < j$, and separate the signs of w_{ij}. For $w_{ij} < 0$, subtracting $2|w_{ij}|$ gives the displayed form. Nonnegativity follows from $-1 \leq \Re(z_i \overline{z}_j) \leq 1$. □

The one-point argument proves the first line after subtraction of the *complete* D_C is positive. It follows that any selected subset of the displayed surplus terms is a valid lower bound for H_C.

The implementation selects only constant–head pairs $(0, n)$, $2 \leq n \leq 1024$. All other surplus terms are left at their proven lower bound zero. In particular, no omitted negative coefficient is discarded: its entire worst-case debit remains in the one-point cone certificate.

The denominators at the two probes need not be equal. Pointwise,

$$\frac{H_C}{2|Q|^2} \geq \frac{\text{selected nonnegative surplus}}{2(1+A_+)^2}. \tag{3}$$

The direction in (3) uses nonnegativity. Averaging the two quotients first and then replacing them by a quotient of averages would in general be false.

4 The translated-phase inequality

For any unit complex numbers z, u,

$$\frac{\Re(zu) + \Re(z\overline{u})}{2} = \Re z\, \Re u.$$

Lemma 3 (Paired-phase lower bound)**.** *For $|z| = |u| = 1$, $\epsilon \in \{-1, 1\}$ and $c_+, c_- \geq c_0 \geq 0$, the following inequality holds.*

$$\frac{c_+[1 - \epsilon\Re(zu)] + c_-[1 - \epsilon\Re(z\overline{u})]}{2} \geq c_0(1 - |\Re u|). \tag{4}$$

This proves a deterministic gain with arbitrary common centre phase. There is no probability, generic-phase or averaging-in-frequency assumption.

At the actual two points $x \pm a$, the first-head phases obey exactly

$$z_n^{\pm} = z_n^c e^{\pm ia \log n/2}. \tag{5}$$

This is an identity for the specified leading source. It continues to hold when the physical truncations at the two points differ; every selected index is at most 1024, far below both truncations. The nonlinear source replacement errors have already been included separately at both physical points in (1). No shared phase is imposed on the gamma or high-tail labels.

Let the two points lie in source frequency rectangles j, k, with certified cone constants C_j, C_k, amplitude caps A_j, A_k, and strict physical floors ℓ_j, ℓ_k. Let $\underline{a}_{j,n}$ bound the first-head amplitude below on rectangle j. Define

$$g_{jk,n} = \max\{0, \log n - 2\max(C_j, C_k), 2\min(C_j, C_k) - \log n\},$$

$$m_{jk,n} = g_{jk,n} \min\left\{\frac{\underline{a}_{j,n}}{2(1 + A_j)^2}, \frac{\underline{a}_{k,n}}{2(1 + A_k)^2}\right\}. \tag{6}$$

If $\log n$ lies between the two thresholds, (6) is zero and no credit is taken. Otherwise both surplus terms in (2) have the same sign convention; (3)–(5) give the certified moment-cone contribution

$$\boxed{G_{jk}^{\mathrm{cone}}(a) = \sum_{n=2}^{1024} m_{jk,n}\left(1 - \left|\cos\frac{a\log n}{2}\right|\right).} \tag{7}$$

This includes recoverable surplus from negative-budget pairs as well as positive pairs. It does not reduce the verified one-point budget D_+.

5 A second gain from the finite Euler logarithmic derivative

Put

$$\rho_p = b_t(p)p^{-[(1+h)/2 + tL/4]}, \qquad \vartheta_p = (x/2 + \pi t/8)\log p.$$

Since $\rho_p < 1$, absolutely convergent expansion gives

$$\Re\frac{E_h}{E} = \frac{1}{2}\sum_{p=2,3,5} \log p \sum_{r\geq 1} \rho_p^r \cos(r\vartheta_p).$$

The complete phase-free cap minus this expression is the nonnegative series

$$\frac{1}{2}\sum_{p} \log p \sum_{r\geq 1} \rho_p^r[1 - \cos(r\vartheta_p)]. \tag{8}$$

The certified Euler cap is at least this full phase-free cap at every source. Applying (4), keeping only the first eight harmonics, gives

$$\boxed{G_{jk}^{\mathrm{Euler}}(a) = \frac{1}{2} \sum_{p=2,3,5} \log p \sum_{r=1}^{8} \min(\underline{\rho}_{j,p}^{r}, \underline{\rho}_{k,p}^{r}) \left(1 - \left|\cos \frac{ra \log p}{2}\right|\right).} \tag{9}$$

All higher harmonics remain covered: their contributions to (8) are nonnegative. They are not bounded by an unjustified zero remainder.

The cone and Euler gains may be added. They arise from two different explicit addends in (1); no common coefficient resource is allocated twice inside one identity. The underlying phases can be dependent–the sum of two valid lower bounds requires no independence.

Theorem 4 (Uniform two-probe improvement)**.** *Assume the complete one-point bounds on source rectangles j, k stated in Section 2. With the lower coefficients in (6)–(9), the two specified translated sources satisfy*

$$\boxed{\frac{S(x+a,h,t) + S(x-a,h,t)}{2} > \frac{\ell_j + \ell_k}{2} + G_{jk}^{\mathrm{cone}}(a) + G_{jk}^{\mathrm{Euler}}(a).} \tag{10}$$

Proof. Apply Lemma 3 to the selected terms in Lemma 2 after the pointwise denominator estimate (3). Apply the same inequality to each of the retained Euler harmonics in (8), and add the two lower bounds in (1). This gives (10). □

All physical approximation and derivative errors are unchanged from the complete one-point inputs. In particular, (10) does not assert that either individual probe satisfies the improved average floor.

6 Uniform lower coefficient bounds and complete spatial coverage

The raw Euler-product head coefficient is

$$c_n(t) = \sum_{d \mid (n,30)} \mu(d) \left(\prod_{p \mid d} b_t(p)\right) b_t(n/d).$$

For $n \le N_0$ it is nonnegative and nondecreasing in $t \ge 0$. Indeed the finite heat graph expansion retains precisely graphs in which all marked selected-prime occurrences are nonisolated. Every retained weight and its heat factor is nonnegative and nondecreasing. Repeated prime occurrences are included. At the three selected primes the coefficient is identically zero.

For a time cell $[t_L, t_R]$ and finite frequency rectangle with upper bound L_j^+, valid lower bounds are

$$\underline{a}_{j,n} = c_n(t_L) \exp\left[-\left(\frac{1+h}{2} + \frac{t_R L_j^+}{4}\right) \log n\right], \tag{11}$$

$$\underline{\rho}_{j,p} = \exp\left[-\frac{1+h}{2} \log p + \frac{t_R}{4} \log p (\log p - L_j^+)\right]. \tag{12}$$

Equation (11) does not assume monotonicity of the product of these factors; it bounds the increasing coefficient and decreasing exponential separately. For (12), $L > \log p$, so the exact weight decreases in both t and L. On the infinite last rectangle, use lower bound zero. Its full previously proved source floor, including every future state activation, is retained.

The source frequency variable is $s = L/2 - \log N_0$, with endpoints

$$0,\ 1/20,\ 3/20,\ 2/5,\ 1,\ 2,\ \infty.$$

Both translated arguments satisfy $x \pm a \geq X - 2$. The calculation checks $X - 4 > 4\pi N_0^2$, so they are in the declared source range. The complete distance between them is less than four, whereas every finite frequency interval has physical width larger than 10^{11}. Thus the possible ordered pairs are exactly covered by

$$(j, j),\ 0 \leq j \leq 5, \qquad (j, j+1),\ 0 \leq j \leq 4. \tag{13}$$

The lower bound used in zero motion is the minimum of (10) over all eleven possibilities. This includes points arbitrarily close to a frequency seam, not just centres away from seams. Physical truncation changes require no additional correlation assumption: the selected low labels occur on both sides, while all other terms and complete state-pair errors remain in the individual source certificates.

7 Variable displacement and the retained kernel laws

For each catalogue displacement $a = rh$, the new gains are computed at that exact rational displacement. There are ten distinct ratios in the retained thirteen-kernel catalogue.

For the sharp continuum law

$$a(y, h) = \sqrt{(h^2 - y^2)/3}, \qquad v = y^2/h^2, \qquad 29v^2 - v - 1 \leq 0,$$

the new supplier uses one conservative displacement interval

$$h\sqrt{(1 - 51/250)/3} \leq a \leq h/\sqrt{3}. \tag{14}$$

The exact positive root v_* is less than $51/250$. Thus (14) contains every displacement at which the sharp kernel can be selected. Each factor $1 - |\cos(a\lambda/2)|$ is replaced by its interval lower bound on (14). The bound includes interior extrema of cosine; it is not interpolation between endpoint cosines. The new calculation is deliberately not optimised over a smaller attained displacement interval.

All displacements are less than two, within the spatial padding in Analytic Foundations, Part A. They are held fixed when differentiating the probe in height. Their possible functional dependence on the hypothetical contact height creates no extra derivative term: the zero-motion comparison evaluates the kernel inequality instantaneously and does not differentiate a moving probe path.

8 Coupling the average to the global zero barrier

For a highest simple upper zero of height y, write $q = y^2$. The retained kernel comparison and the complete own-pair subtraction give

$$q' \leq -2 - \frac{2\lambda q}{h}\left[\frac{S(x + a, h, t) + S(x - a, h, t)}{2} - S_{\mathrm{self}}(q)\right]. \tag{15}$$

Its exact self-pair contribution is

$$S_{\mathrm{self}}(q) = \frac{2h(A + h^2 - q)}{(A + h^2 - q)^2 + 4Aq}, \qquad A = a^2.$$

For the sharp continuum kernel, $\lambda = 2$ and

$$S_{\mathrm{self}}(q) = \frac{6}{h(4 - q/h^2)}.$$

No self-pair credit is taken twice. A stronger bound for the average in (10) directly strengthens the coefficient in (15).

On every comparison step, the exact consumer chooses only a kernel whose height-ratio restriction is satisfied by the upward enclosure of q. The source floor is the matching *joint* floor for that displacement, not an individually improved probe floor. The self-pair upper bound is checked on the entire permitted decreasing range of q. Thus a constant rate $c > 0$ gives $q' < -2 - cq$ through the step.

The source schedule has 608 cells of width $1/4000$, with unchanged rational probe heights. The archived joint-probe baseline used four comparison steps per cell. The publication refinement uses 128 steps per cell, as proved and checked in Section 11. The initial bound is $q_0 = (1.000001)^2 > 1$; the final positive height is $q_e = 10^{-8}$, followed by de Bruijn contraction costing $q_e/2$. Upward dyadic enclosures of each exponential solution are used. A distinct rational exponential/logarithm implementation supplies a second endpoint.

The global first-contact argument is given in Analytic Foundations, Part A, Section 11, using the zero-dynamics and confinement results of [1, Proposition 3.1 and Theorem 1.5(i)]: the wall excludes nonreal zeros from the finite head, positive-time confinement makes first contact attainable, backward Hermite splitting excludes multiple contact, and the simple-zero inequality contradicts first contact. At a time knot, the left-hand row is used. The initial strict margin is checked numerically, so compactness is used only after positive time. The proof does not infer a smaller-time strip by reversing heat flow.

9 Numerical verification and reproducibility

The new arithmetic lower bounds use two constructions of every raw head coefficient through 1024: complete Euler-divisor inclusion–exclusion, and descending multiplication by the three finite Euler factors. All subsequent transcendental values and cosine extrema are enclosed with Arb/FLINT[7] at 192 and 256 bits. Coefficients, denominator reciprocals and attenuation factors are rounded down on the 2^{-80} grid. Their sums are then calculated with exact Python integers. Zero may replace a negative lower endpoint only where the analytic positivity theorem has established it.

The two lanes produce the same finite-grid hashes. Final acceptance retains the weaker of the two joint lower bounds. Both new lanes use the same source-row selection from independently validated one-point rows: the stronger valid complete enclosure can be selected at each matching rectangle. This is not a claim that the coarser one-point lane independently proves the final headline. Both full one-point lanes, their remainders and all domain conditions are regenerated and checked.

All eleven spatial pair classes, all 608 time cells and all eleven translation modes are required. The last spatial interval is infinite and retains its full old floor even though its additional lower gains are zero. The exact checker reconstructs all scalar gain sums, matching domains and minima. With four steps per cell and the gains disabled, the archived baseline reproduces the exact preceding endpoint. Its separate cone-only and Euler-only comparisons measure the effects without pretending their endpoint improvements add linearly. The final release changes only the scalar subdivision, not these source gains.

Supplementary direct paired checks include an ordinary source, a frequency seam and a physical truncation seam. They use the actual nonlinear source coordinates, full two-branch

sums, actual cutoffs and complete derivative error. They are not acceptance premises for the uniform source theorem.

The replay checks integrity and regenerated numerical inequalities. It does not formally verify the infinite analytic implications or reprove the published finite-RH computation. Neither of the new lanes is an independent transcendental library. Exact identities, interval rounding and the scalar ODE enclosure still depend on their stated arithmetic implementations.

10 Contribution and limitations

The new source gain is not obtained from a favourable physical sample or a conjecture that two worst cases are unlikely. It follows from the exact relation between selected phases at two specified translations and from nonnegative surplus terms already present in the complete one-point proof. The physical denominators may differ. The gamma phase and high branch phases are not synchronised by assumption. All their previous errors remain.

No product of two million-term polynomials is expanded. Conversely, this is not the optimal joint two-point estimate: only constant–head pairs through 1024 and eight harmonics of each selected Euler factor are used. Correlation among other head labels, the reflected branch and the two denominators is not credited beyond the already proved one-point bounds. Enlarging the selected positive-surplus set can only help the analytic lower bound, but whether its certified numerical gain is worthwhile must be evaluated.

The time–height schedule and kernel catalogue are not globally optimised. The result does not establish $\Lambda < 0.09$ or RH, and no exhaustive world-record or priority claim is made. It is a further checked corollary obtained from the analytic estimates and numerical certificates supplied here, with a uniform two-point improvement.

11 The exact terminal comparison

The only final mathematical change to the archived $\Lambda < 0.149399$ calculation is the number of comparison substeps. The schedule, all source floors, the selected head indices through 1024, the eight Euler harmonics, the kernel catalogue, the spatial padding, the self-pair subtraction and all physical-error contracts are unchanged.

Every source lower bound is valid throughout its full time cell. At each new substep, let q be an upward enclosure of the ideal comparison height. Only a kernel valid on the entire interval from zero to that height is admissible. Its matching translated-source floor and complete self-pair cap give a rational $c > 0$ with $q' < -2 - cq$ throughout the substep. Refining the time partition therefore requires no new functional hypothesis.

For a substep of width Δt, the exact solution is

$$q_{\mathrm{next}} = (q + 2/c)e^{-c\Delta t} - 2/c.$$

One construction uses the even alternating Taylor upper bound through degree 18 and 96-bit upward dyadic rounding. The second uses the reciprocal of the positive exponential Taylor sum through degree 12 and 112-bit upward rounding. Both enclose the same continuous comparison, not a discretely jumping zero trajectory.

If the step reaches $q_e = 10^{-8}$, put

$$z = \frac{c(q - q_e)}{2 + cq_e}.$$

The remaining time is $\log(1+z)/c$. Each construction bounds this logarithm upward, checks that the terminal time remains inside the certified source cell, adds $q_e/2$, and rounds the final time upward.

The fixed-source audit was:

Substeps per source cell	Primary exact upper time, displayed	Alternative exact upper time, displayed
4	0.149398159734722	0.149398159735734
8	0.149396459831885	0.149396459832690
16	0.149395675207179	0.149395675207190
32	0.149395321833941	0.149395321833950
64	0.149395152324926	0.149395152324926
128	0.149395073827688	0.149395073827688

The final two constructions, at 128 substeps per source cell, both give

$$\boxed{U = \frac{18674384228461}{125000000000000} = 0.149395073827688.}$$

Compared with the previous exact value 0.149398159734722, the decrease is exactly

$$0.000003085907034.$$

These are exact comparisons, not floating-point optimisation scores.

A separate history checker reconstructs every rational rate and time join, verifies the full quadratic self-pair inequality on the permitted height interval, and checks each upward endpoint with an even degree-20 exponential bound. It independently validates terminal logarithms with the odd degree-13 upper series. It rejects modified source floors, rates, time joins, endpoint ceilings and terminal lags.

11.1 Scope of the numerical refinement

This is a bounded refinement of the scalar consumer of already established functional estimates. It is not an exhaustive optimisation over probe heights, kernel families, selected surplus terms, or source schedules. Further such changes may improve the bound, but require their own complete verification.

The release is frozen at the safe public ceiling $\Lambda < 0.149396$. No optimality or worldwide priority claim follows from this audit. The number of internal computational iterations is not the publication version: this separate paper's first prepared public release is version 1.0.

12 Data availability and author disclosure

The deposited materials comprise this manuscript, Analytic Foundations, the numerical programs and runtime, the complete interval enclosures, exact comparison histories, verification instructions and integrity manifests. The default regeneration command is given in the Verification Guide. It recomputes the numerical inequalities; an integrity-only command checks file identities and does not verify the mathematics. The scientific evidence archive is unchanged by this editorial revision.

The numerical dependencies are identified by content hashes in the deposit. The one-point source and sharp-geometry proofs are reproduced in Analytic Foundations rather than treated as

inaccessible external assumptions. The mathematical inputs from other authors are the published results cited in the dependency table in Section 1. Neither [4] nor [6] supplies a theorem, estimate or numerical certificate used to obtain the present bound.

Author disclosure. AI assistance was used in mathematical exploration, programming, numerical experimentation, checking and drafting. The author takes responsibility for the claims, their presentation and the deposit. Automated review and numerical replay are not external peer review. No complete proof-assistant formalisation is claimed. The earlier and concurrent work cited in the introduction was considered in preparing the comparison and exposition; independence of proof inputs is not a claim of historical priority or of absence of awareness of that work.

Publication status. This is a preprint prepared for independent mathematical and computational review. No journal acceptance or exhaustive record-ranking claim is made. The DOI assigned to this separate record should be used for this manuscript; the DOI of the earlier paper in [5] is not its identifier.

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